

HCI 5840 assignment

Important: This is the original sketch prior to instructor feedback to simplify. Please see project specs for updated version.

Title: Canvas Backup/Alignment Prep

Overview

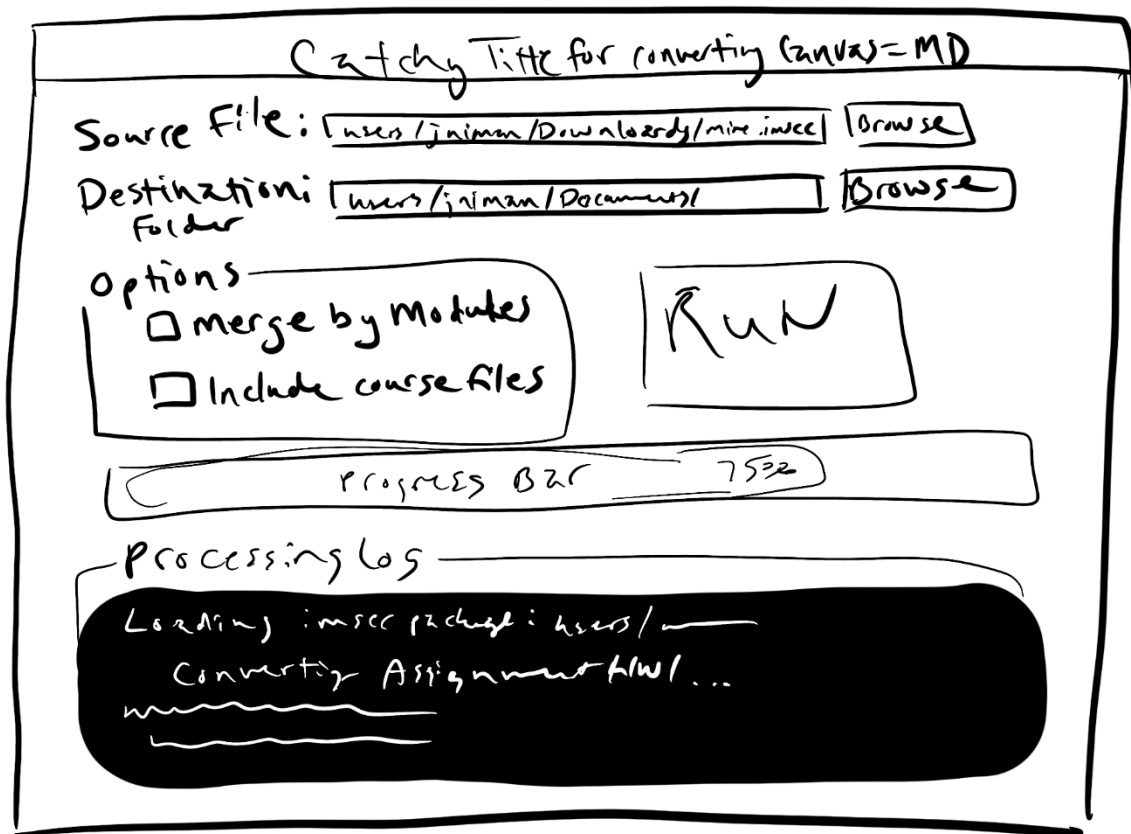
I'm not sure what catchy title to call this, but essentially, I'd like to build a desktop application (using Tkinter) that will extract, organize, and convert a Canvas course export package (.imsc, which compresses XML, html, and other course files) into logically named and structured Markdown files mapped by Canvas designated modules. Canvas exports use nonsensical (to humans) file naming conventions which makes it difficult for us to find and organize the offline Canvas content outside of the LMS (learning management system).

This tool will convert and organize the Canvas content into markdown so the files can be uploaded directly to NotebookLM for analysis to help prepare instructors and instructional designers (primary stakeholders) and Quality Matters reviewers (secondary stakeholders) for alignment analysis.

What problem would it (help to) solve?

Instructors going through our Quality Matters (QM) course certification process have expressed interest in using generative AI to help analyze alignment between course materials, assessments, activities, and their learning objectives. We need a method to pull everything out of Canvas and add it to NotebookLM, which is an AI tool that would ensure only the course materials are utilized as resources (without influence from external factors).

Sketch



User path/interaction/processing/vignette

1. An instructional designer (ID) is assisting an instructor who has limited time and resources to thoroughly review QM standards and would like help prioritizing next steps.
2. The ID is not a content expert, so they ask permission to use an ISU approved tool, NotebookLM, to set up an environment for analyzing standards and asking questions about course content.
3. After receiving permission, the ID (user) starts up [insert catchy title here].exe
4. The ID clicks "browse" and selects the course's .imsc file through the desktop (Tkinter) GUI.
5. Optional: They could select to merge Canvas content into modules and pull the course files into the same destination directory
6. The ID can specify the destination directory, then press "run"
7. **Data/processing:** The program unzips package, parses the .xml via BeautifulSoup, extract HTML content, convert via markdownify, structure by module for naming and possible merging (more later, in feasibility).
8. **Results:** This outputs a folder containing logically labeled markdown documents of the Canvas content (pages, assignments, etc.), original files (PPT, docx, PDF), and an export summary log.

9. The user can inspect the output log to see if they need to troubleshoot anything within Canvas (missing files, incorrectly mapped modules), in case they need to fix something and re-export.
10. Markdown files are ready to be uploaded to NotebookLM. They can choose to also upload the long prompts I've prepared as markdown files (I assume these are stored separately in Github since it's optional?)

Feasibility + challenges/limitations

- Canvas's export package does not include external tools/content
 - This means New Quizzes would need to be manually exported (although they are working on internalizing it so eventually it should be included)
 - Studio, Echo360, or Panopto videos offer transcripts, but they need to be manually downloaded
 - I could provide an additional layer that offers to web scrape these, but I'm not sure how reliable this would be since LTIs may not necessarily have any specific "tells" so I can direct playwright or another scraper
 - For example, videos likely won't have embed codes to look for
- NotebookLM through Iowa State's access limits resources to 50
 - Our access provides us data protection, for instructors worried about AI training off their content
 - A layer to add to address this would be to program the option or ability to merge pages and assignments under each module to reduce the number of resources, ideally, to 15 or 16 (typical number of modules/weeks).
 - Not sure how this would work with course files like powerpoint slides, documents, PDFs. They may just exist outside the merged module markdowns that count against the resource limit since I can't imagine a clean conversion for PowerPoint, as it has so many layers. PDFs are also iffy if they are not appropriately tagged, but AI may be able to read them as is.

Self vs help

I'll need to go back to my HCI 5740 notes on BeautifulSoup and XML parsing, path, and glob. I feel like this is work I can do through review. I will need to find additional modules to help with the rest (like markdownify). I think this project is the most doable for me since I'm still very fresh to Python, but it can be scalable as well. I can try to troubleshoot ways to find and merge specific files, or potentially add a web scraping option to find the external content, though I imagine that I'd need more help with web scraping.

This is my preferred project because instructional designers in my department could start using it right away. One other thing to note is that I'm choosing this over an API key because those are pretty locked down at our institution and this is something I'd like to actively use and support in our office with live courses (and the instructor's permission/interest).